



AQ3.4: Activity Questions 4 - Not Graded



This assignment will not be graded and is only for practice.

Level 1

Consider the following statements:

- Statement 1: $0(0, 0) + 0(1, 1) = (0, 0)$ implies that the set $\{(0, 0), (1, 1)\}$ is a linearly independent subset of R^2 .
- Statement 2: $2(1, 0) + 2(0, 1) = (2, 2)$ implies that the set $\{(1, 0), (0, 1), (2, 2)\}$ is a linearly dependent subset of R^2 .
- Statement 3: $2(1, 0) + 2(0, 1) = (2, 2)$ implies that the set $\{(1, 0), (2, 2)\}$ is a linearly dependent subset of R^2 .

Find the number of correct statements.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 1

1 point

1 point

If $\alpha(1, 1, 0) + \beta(0, 1, 1) + \gamma(1, 0, 1) = \nu$ for some $\alpha, \beta, \gamma \in R$ and $\nu \in R^3$, then choose the set of correct options.

☐

If $\nu = (0, 0, 0)$, then one of the possible solutions of α, β , and γ is $\alpha = \frac{1}{2} = \gamma$ and $\beta = -\frac{1}{2}$.

☐

If $\nu = (0, 0, 0)$, then one of the possible solutions of α, β , and γ is $\alpha = \beta = \gamma = 0$.

☐

If $\nu = (0, 0, 0)$, then the possible solution of α, β , and γ is unique.

☐

If $\nu = (1, 0, 0)$, then one of the possible solutions of α, β , and γ is $\alpha = \frac{1}{2} = \gamma$ and $\beta = -\frac{1}{2}$.

☐

The set $\{(1, 1, 0), (0, 1, 1), (1, 0, 1)\}$ is linearly independent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $\nu = (0, 0, 0)$, then one of the possible solutions of α, β , and γ is $\alpha = \beta = \gamma = 0$.

If $\nu = (0, 0, 0)$, then the possible solution of α, β , and γ is unique.

If $\nu = (1, 0, 0)$, then one of the possible solutions of α, β , and γ is $\alpha = \frac{1}{2} = \gamma$ and $\beta = -\frac{1}{2}$.

The set $\{(1, 1, 0), (0, 1, 1), (1, 0, 1)\}$ is linearly independent.

1 point



Consider a set of vectors $S = \{(1, 2), (-1, 2), (3, 1), (-3, 1)\}$ in R^2 . Choose the set of correct options.

☐

$\{(1, 2), (-1, 2)\}$ is a linearly dependent set.

☐

$\{(-1, 2), (-3, 1)\}$ is a linearly independent set.

☐

$\{(-3, 1), (-1, 2), (3, 1)\}$ is a linearly dependent set.

☐

$\{(3, 1), (-3, 1)\}$ is a linearly dependent set.

☐

$\{(3, 1), (1, 2), (-3, 1)\}$ is a linearly dependent set.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$\{(-1, 2), (-3, 1)\}$ is a linearly independent set.

$\{(-3, 1), (-1, 2), (3, 1)\}$ is a linearly dependent set.

$\{(3, 1), (1, 2), (-3, 1)\}$ is a linearly dependent set.

1 point

Let V be a vector space and non zero vectors v_1, v_2, v_3 , and $v_4 \in V$. If v_1 is a linear combination of v_i for $i = 2, 3, 4$, then which of the following is (are) correct?

☐

The set $\{v_1, v_2, v_3, v_4\}$ cannot be linearly independent.

☐

The set $\{v_2, v_3, v_4\}$ must be linearly dependent.

☐

The set $\{v_2, v_3, v_4\}$ must be linearly independent.

☐

The set $\{v_1\}$ must be linearly independent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The set $\{v_1, v_2, v_3, v_4\}$ cannot be linearly independent.

The set $\{v_1\}$ must be linearly independent.

1 point

Choose the correct set of options.

☐

Union of two distinct linearly independent sets may be linearly independent.

☐

Union of two distinct linearly independent sets must be linearly dependent.

☐

Non-empty intersection of two linearly independent sets must be linearly independent.

☐

Non-empty intersection of two linearly independent sets must be linearly dependent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

Union of two distinct linearly independent sets may be linearly independent.

Non-empty intersection of two linearly independent sets must be linearly independent.

If vectors v_1, v_2, v_3 are linearly independent and $\alpha v_1 + \beta v_2 + \gamma v_3 = 0$, then find the value of $\alpha + \beta + \gamma$.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 0

1 point

Consider the set of matrices $S = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \begin{bmatrix} 7 \\ 8 \\ x \end{bmatrix} \right\} \subset M_{3 \times 1}(R)$ $S = \left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \begin{bmatrix} 7 \\ 8 \\ x \end{bmatrix} \right\} \subset M_{3 \times 1}(R)$

$\left\{ \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}, \begin{bmatrix} 7 \\ 8 \\ x \end{bmatrix} \right\} \subset M_{3 \times 1}(R)$. If $x \in \{9, 4, 2, 0\}$, then find the maximum value of x so that S is linearly independent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

(Type: Numeric) 4

1 point

Level 2

1 point

Let S be a linearly independent subset of R^3 . Choose the correct set of options.

☐

$S \cup \{(1, 0, 0)\}$ must be linearly independent.

☐

$S \cup \{v\}$ must be linearly independent for any $v \in R^3$.

☐

$S \setminus \{v\}$ must be linearly independent for any $v \in R^3$.

☐

There may exist some $v \in R^3$, such that $S \cup \{v\}$ is still linearly independent.

No, the answer is incorrect.

Score: 0

Accepted Answers:

$S \setminus \{v\}$ must be linearly independent for any $v \in R^3$.

There may exist some $v \in R^3$, such that $S \cup \{v\}$ is still linearly independent.

1 point

Consider a set of vectors $S = \{(1, 1, -1), (-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2), (1, 0, -2)\}$. Choose the set of correct options.

☐

The singleton set $\{(0, 1, -2)\}$ is linearly dependent.

☐

If $\alpha, \beta \in S$ and α, β are distinct then $\{\alpha, \beta\}$ is a linearly independent set of vectors.

☐

The set $\{(-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2)\}$ is a linearly independent set of vectors.

☐

The set S is a linearly independent set of vectors.

☐

The set $\{\alpha, \beta, \gamma\}$ is a linearly dependent set of vectors for any $\alpha, \beta, \gamma \in S$, where

all the three are distinct vectors.



The set $\{\alpha, \beta, \gamma, \delta\}$ is a linearly independent set of vectors for any $\alpha, \beta, \gamma, \delta \in S$, where all the four are distinct vectors.

No, the answer is incorrect.

Score: 0

Accepted Answers:

If $\alpha, \beta \in S$, α, β are distinct then $\{\alpha, \beta\}$ is a linearly independent set of vectors.

The set $\{(-1, 1, 1), (0, \frac{1}{2}, 0), (0, 1, -2)\}$ is a linearly independent set of vectors.

1 point

Let SS be the solution set of a system of homogeneous linear equations with 3 variables and 3 equations, whose matrix representation is as follows:

$$Ax = 0$$

where A is the 3×3 coefficient matrix and x denotes the column vector $\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$.

Choose the set of correct options.



If v_1 and v_2 are in SS , then any linear combination of v_1 and v_2 will also be in SS .



The set SS will be a subspace of R^3 , with respect to usual addition and scalar multiplication as in R^3 .



The set $\{v_1, v_2, v_1 - v_2\}$ is a linearly dependent subset in SS .



The set $\{v_1, v_2\}$ is a linearly independent subset in SS if v_1 is not a scalar multiple of v_2 .

No, the answer is incorrect.

Score: 0

Accepted Answers:

If v_1 and v_2 are in SS , then any linear combination of v_1 and v_2 will also be in SS .

The set SS will be a subspace of R^3 , with respect to usual addition and scalar multiplication as in R^3 .

The set $\{v_1, v_2, v_1 - v_2\}$ is a linearly dependent subset in SS .

The set $\{v_1, v_2\}$ is a linearly independent subset in SS if v_1 is not a scalar multiple of v_2 .

[Check Answers](#)

Your score is: 0/10